



TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CIVICS GROUP

H2 CHEMISTRY

9729/03

Paper 3 Free Response

20 September 2023

2 hours

Candidates answer on the Question Paper.

Additional materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and Civics Group in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper. If additional space is required, you should use the pages at the end of this booklet. The question number must be clearly shown.

Section A

Answer all questions.

Section B

Answer **one** question.

A Data Booklet is provided.

The use of an approved calculator is expected, where appropriate.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 18
2	/ 19
3	/ 23
Section B	
4	/ 20
OR	
5	/ 20
Total	

Section A

Answer **all** the questions in this section.

1 The Group 17 elements and its halides can react with many transition metals.

(a) (i) Use the *Data Booklet* to describe the relative reactivity of the Group 17 elements as oxidising agents and relate this reactivity to relevant $E^\ominus$ values. [2]

(ii) The following reagents are contained in three bottles labelled A, B and C.

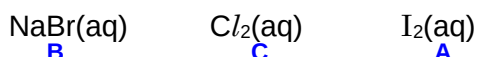


Table 1.1 shows the results of tests that were carried out using the reagents in the bottles. In each test, equal volumes of the reagents were mixed and the resultant colour of the solution was recorded.

Table 1.1

test	reagents mixed	colour of resultant solution
1	A and B	brown
2	C and B	orange
3	A and C	brown

State the identity of the reagent in bottle A and write equations for any reactions that occur. [2]

(i) Down the group, $E^\ominus$ becomes less positive (trend or quote values). Hence, the oxidising power of the halogens decreases down the group. (or lower tendency for the halogen (X_2) to gain electrons/be reduced.)

(ii) The reagent in bottle A is $\text{I}_2\text{(aq)}$.



- (b) Cobalt forms many complexes with ligands such as H_2O , NH_3 and Cl^- . The most common oxidation states of cobalt are +2 and +3.

Table 1.2 gives information about two octahedral cobalt(III) complexes **D** and **E**.

Table 1.2

	complex	colour of aqueous ion
D	$[\text{CoCl}_2(\text{NH}_3)_4]^+$	violet
E	$[\text{Co}(\text{NH}_3)_6]^{3+}$	yellow

- (i) Define the term *complex*. [1]
- (ii) The five d orbitals in a gas-phase transition metal ion are degenerate, but are split into two levels when the ion is in an octahedral complex.
- Explain why the d orbitals are split into two levels in an octahedral complex, including which of the d orbitals are in each level. [2]
- (iii) Draw the shapes of the d orbitals that occupy the higher energy level. [2]
- (iv) The colour of the complexes is due to the absorption of light at specific wavelengths. The colour observed is the complement of the colour absorbed.

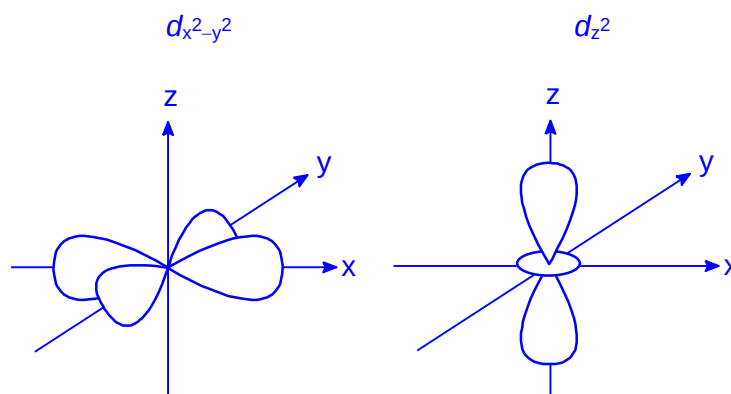
Suggest why the two complexes **D** and **E** have different colours. [1]

(i) A transition metal complex contains a central metal atom/ion attached to ligands through dative / coordinate bonds.

(ii) The $d_{x^2-y^2}$ and d_{z^2} orbitals have their lobes pointing at the ligands along the axes. Hence, they experience greater repulsion from the ligands. On the other hand, the d_{xy} , d_{xz} and d_{yz} orbitals experience less repulsion as their lobes are in between the coordinate axes.

As a result, the five d orbitals are split into 2 energy levels, with the $d_{x^2-y^2}$ and d_{z^2} orbitals having the higher energy level and the d_{xy} , d_{xz} and d_{yz} orbitals having the lower energy level.

(iii)

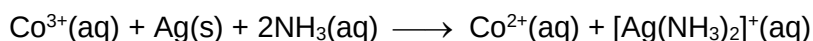


(iv) Cl^- and NH_3 have different ligand (field strength), resulting in different size of energy gap.

- (c) An electrochemical cell is set up using a $\text{Co}^{3+}(\text{aq})/\text{Co}^{2+}(\text{aq})$ half-cell and a $[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})/\text{Ag}(\text{s})$ half-cell under standard conditions.

The electrode potential of the $[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})/\text{Ag}(\text{s})$ half-cell is -0.37 V .

When the cell discharges, the following reaction occurs.



- (i) Write a half-equation for the oxidation of Ag to $[\text{Ag}(\text{NH}_3)_2]^+$. [1]
- (ii) With the aid of the *Data Booklet*, calculate the $E_{\text{cell}}^\ominus$ and $\Delta G^\ominus$ for the above electrochemical reaction. [2]
- (iii) Explain the effect on $E_{\text{cell}}^\ominus$ when H^+ is added to the $[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})/\text{Ag}(\text{s})$ half-cell. [2]



- (iv) The Nernst equation can be used to determine the cell potential under non-standard conditions and is given as follows:

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{0.0592}{n} \log_{10} Q$$

where the value of Q is calculated using this expression $\frac{[\text{Co}^{2+}][\text{Ag}(\text{NH}_3)_2^+]}{[\text{Co}^{3+}][\text{NH}_3]^2}$

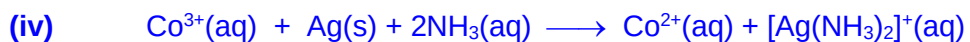
and n is the number of moles of electrons transferred per mole reaction.

When the electrochemical cell described in (c) discharges, the concentration of Co^{3+} in the $\text{Co}^{3+}(\text{aq})/\text{Co}^{2+}(\text{aq})$ half-cell decreases from 1.00 mol dm^{-3} to 0.75 mol dm^{-3} .

Show that the value of Q is 8.33 when $[\text{Co}^{3+}]$ is 0.75 mol dm^{-3} .

Hence, calculate the cell potential at this Co^{3+} concentration. [3]

- (i) Overall (given): $\text{Co}^{3+} + \text{Ag}(\text{s}) + 2\text{NH}_3 \longrightarrow \text{Co}^{2+} + [\text{Ag}(\text{NH}_3)_2]^+$
 Oxidation: $\text{Ag}(\text{s}) + 2\text{NH}_3 \longrightarrow [\text{Ag}(\text{NH}_3)_2]^+ + \text{e}^-$
- (ii) $E_{\text{cell}}^{\ominus} = +1.89 - (-0.37) = \underline{+2.26 \text{ V}}$
 $\Delta G^{\ominus} = -nFE_{\text{cell}}^{\ominus} = 1 \times 96500 \times 2.26 = \underline{-2.18 \times 10^5 \text{ J mol}^{-1}}$
- (iii) $[\text{Ag}(\text{NH}_3)_2]^+ + \text{e}^- \rightleftharpoons \text{Ag}(\text{s}) + 2\text{NH}_3 \quad E_{\text{ox}}^{\ominus} = -0.37 \text{ V}$
 H^+ will react with NH_3 in the half cell causing $[\text{NH}_3]$ to decrease. By Le Chatelier's Principle, the equilibrium position will shift right to increase $[\text{NH}_3]$ and $E_{\text{ox}}^{\ominus}$ becomes more positive. As a result, $E_{\text{cell}}^{\ominus}$ will become less positive.



Initial	1.00		1.00	1.00	1.00
Change	-0.25		-2(0.25)	+0.25	+0.25
Final	0.75		0.50	1.25	1.25

When $[\text{Co}^{3+}] = 0.75 \text{ mol dm}^{-3}$,
 $[\text{NH}_3] = 0.50$, $[\text{Co}^{2+}] = [\text{Ag}(\text{NH}_3)_2]^+ = 1.25 \text{ mol dm}^{-3}$

$$Q = \frac{[\text{Co}^{2+}][\text{Ag}(\text{NH}_3)_2^+]}{[\text{Co}^{3+}][\text{NH}_3]^2} = \frac{1.25 \times 1.25}{0.75 \times (0.5)^2} = \underline{\underline{8.33 \text{ (shown)}}}$$

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cell}}^{\ominus} - \frac{0.0592}{n} \log_{10} Q \\ &= 2.26 - \frac{0.0592}{1} \log_{10} 8.33 \\ &= \underline{\underline{+2.21 \text{ V}}} \end{aligned}$$

[Total: 18]



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2 Inorganic compounds have various uses in the food and medical industry. For example, sodium hydroxide is used in the chemical peeling of fruits and vegetables, whereas magnesium compounds are used as food additives.

- (a) Sodium hydroxide is formed in an explosive reaction between sodium metal and water, as shown in *reaction 1* in Table 2.1. The table also shows some other equations and their related standard enthalpy changes.

Table 2.1

		standard enthalpy change / kJ mol ⁻¹
reaction 1	$\text{Na(s)} + \text{H}_2\text{O(l)} \longrightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq}) + \frac{1}{2}\text{H}_2(\text{g})$	$\Delta H_1^\ominus$
reaction 2	$\text{Na(s)} \longrightarrow \text{Na(g)}$	+107
reaction 3	$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \longrightarrow \text{H}_2\text{O(l)}$	-58
reaction 4	$\text{Na}^+(\text{g}) + \text{e}^- + \text{H}^+(\text{aq}) \longrightarrow \text{Na}^+(\text{aq}) + \frac{1}{2}\text{H}_2(\text{g})$	-850

- (i) Deduce the sign of $\Delta S^\ominus$ for reaction 1. [1]
- (ii) Hence, predict how an increase in temperature affects the spontaneity of reaction 1. Explain your answer. [1]
- (iii) Write an equation to represent the first ionisation energy of sodium. [1]
- (iv) Use the data in Table 2.1 and relevant data from the *Data Booklet* to draw an energy cycle to calculate the standard enthalpy change of reaction 1, $\Delta H_1^\ominus$. [3]

(i) $\Delta S^\ominus$ is **positive** because there is **increase in the moles of gaseous particles (or there are mobile ions in aqueous solution)**, leading to **greater degree of disorder/increase in entropy**.

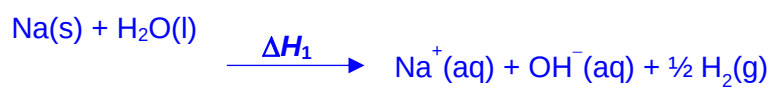
(ii) $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$

Since $\Delta H < 0$ and $\Delta S > 0$, **ΔG becomes more negative as temperature increases**. Therefore reaction 1 becomes **more spontaneous as temperature increases**.

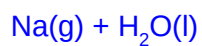
(iii) $\text{Na(g)} \rightarrow \text{Na}^+(\text{g}) + \text{e}^-$



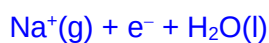
(iv)



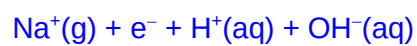
+107



+494

(1st IE of Na)

-850



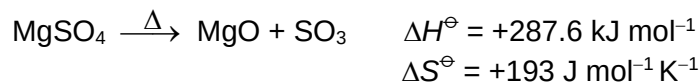
-58

By Hess' Law,

$$+107 + 494 + (-850) = \Delta H_1 + (-58)$$

$$\Delta H_1 = \underline{\underline{-191 \text{ kJ mol}^{-1}}}$$

- (b) Magnesium sulfate is used as an additive to enhance the firmness of canned vegetables. It decomposes into magnesium oxide and sulfur trioxide according to the following equation.



- (i) Calculate $\Delta G^\ominus$ for the above reaction. [1]
- (ii) Calculate the theoretical minimum temperature at which decomposition of MgSO_4 starts to occur. [2]
- (iii) It has been found experimentally that magnesium sulfate starts to decompose at 1124°C . Suggest a possible reason for the discrepancy between your calculated value in (ii) and the experimental value provided. [1]
- (iv) The product of the above decomposition, magnesium oxide, is a white hygroscopic solid with a melting point 2850°C .

Draw a labelled diagram to illustrate the arrangement of the lattice particles in solid magnesium oxide. Your diagram should show clearly the relative size of the different types of particles. [1]

- (v) Magnesium oxide and silicon dioxide are both used as lining for furnaces. Using structure and bonding, explain why the melting point of magnesium oxide is higher than that of silicon dioxide. [3]

(i) $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$
 $= +287.6 - (298)(+193/1000) = \underline{+230 \text{ kJ mol}^{-1}}$

(ii) For reaction to be spontaneous, $\Delta G < 0$

$$\Delta G = \Delta H - T\Delta S < 0$$

$$\Delta H < T\Delta S$$

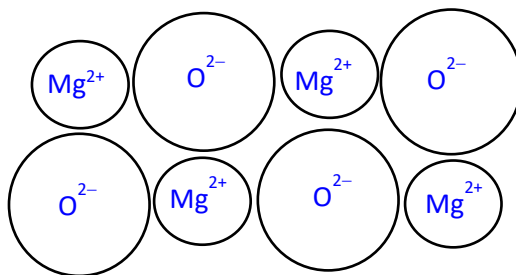
$$+287.6 < T \left(\frac{193}{1000} \right)$$

$$T > \underline{1490 \text{ K}} \text{ (or } 1220^\circ\text{C (3sf))}$$

(iii) Value obtained in (ii) was based on the assumption that entropy and enthalpy changes were independent of temperature. (or decomposition reaction does not occur under standard conditions)



(iv)



- (v) Magnesium oxide has a giant ionic lattice structure.
Silicon oxide has a giant molecular structure.
More energy required to overcome the stronger ionic bonds/electrostatic forces of attraction between ions in MgO than the covalent bonds between Si and O atoms in SiO₂.

(c) Magnesium carbonate is sparingly soluble in water. It is therefore commonly used as an additive in table salt to prevent it from clumping. Magnesium carbonate has a solubility of $1.26 \times 10^{-4} \text{ mol dm}^{-3}$ at 25°C .

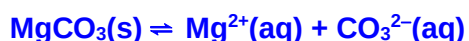
- (i) Write the solubility product, K_{sp} , expression for magnesium carbonate, stating its units. [2]
- (ii) Calculate a value for the solubility product, K_{sp} , for magnesium carbonate. [1]
- (iii) When 0.0500 mol of solid Na_2CO_3 is added to a 1.0 dm^3 saturated solution of MgCO_3 at 25°C , a precipitate forms in the solution.

Calculate the concentration of $[\text{Mg}^{2+}]$ in the resultant solution. [2]

(i) $K_{\text{sp}} = [\text{Mg}^{2+}][\text{CO}_3^{2-}] \text{ mol}^2 \text{ dm}^{-6}$

(ii) $K_{\text{sp}} = [\text{Mg}^{2+}][\text{CO}_3^{2-}]$
 $= (1.26 \times 10^{-4})^2$
 $= 1.59 \times 10^{-8} \text{ mol}^2 \text{ dm}^{-6}$

(iii) Let the $[\text{Mg}^{2+}]$ in this resultant solution be $s \text{ mol dm}^{-3}$.



Eqm conc. $s \quad (s + 0.0500)$

$$K_{\text{sp}} = s(s + 0.0500) \approx s(0.0500) \quad [\text{assume } s \ll 0.0500]$$

$$1.59 \times 10^{-8} = s(0.0500)$$

$$[\text{Mg}^{2+}] = s = 3.18 \times 10^{-7} \text{ mol dm}^{-3}$$

[Total: 19]



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- 3 (a) Remifentanyl ($C_{20}H_{28}N_2O_5$) is a synthetic opioid drug. It is given to patients during surgery to relieve pain and as an adjunct to an anesthetic.

A possible synthesis route of Remifentanyl is shown in Fig. 3.1.

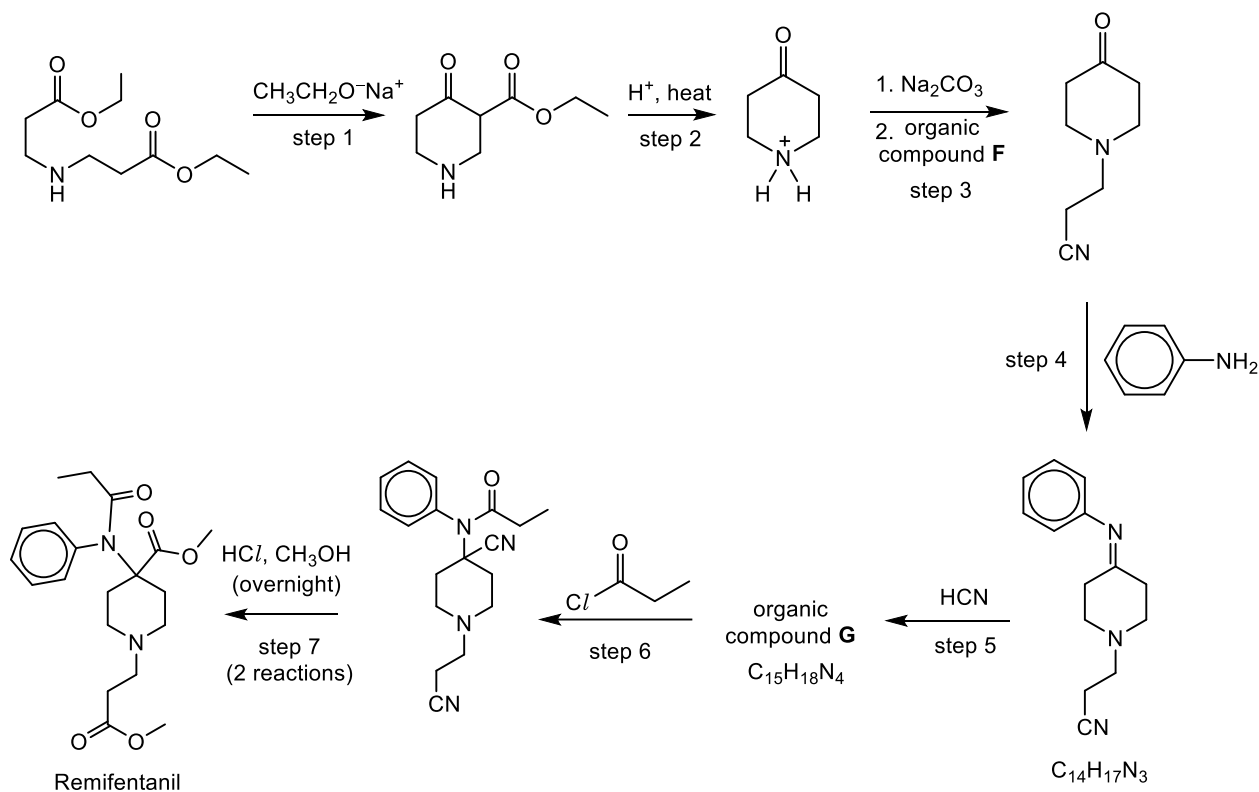
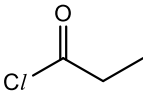
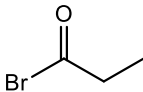


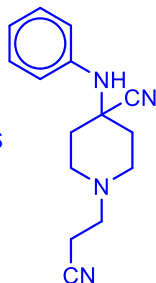
Fig. 3.1

- (i) Name the types of reactions occurring in step 4 and step 7 (2 reactions occur in step 7). [3]
- (ii) Draw the structure for the organic compound **G**. [1]
- (iii) Suggest why step 5 is carried out at a temperature of about $10^\circ C$. [1]
- (iv) When the reagent  in step 6 is replaced by , it was found that the reaction was five times faster. Suggest an explanation for this observation. [1]
- (v) Given that step 3 is a nucleophilic substitution reaction, suggest:
- why sodium carbonate was used and
 - the structure of the organic compound **F**. [2]

(i) Step 4: condensation

Step 7: acidic hydrolysis
condensation or nucleophilic (acyl) substitution

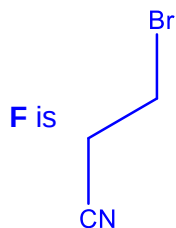
(ii) **G** is



(iii) HCN is a toxic gas with a low boiling point.

(iv) C-Br is weaker than C-Cl, thus less energy is required to break the bond.

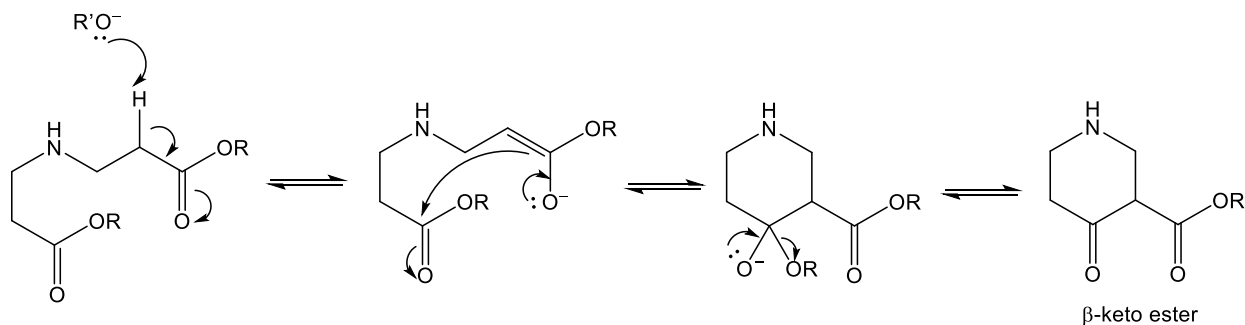
(iv) Na_2CO_3 was added to neutralise the acidic salt/cation to form the amine/nucleophile.



(b) Step 1 of the reaction scheme in Fig. 3.1 involves the Dieckmann condensation.

This intramolecular condensation reaction occurs between 2 ester groups to give a cyclic β -keto ester.

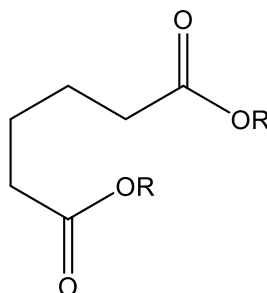
Fig. 3.2 shows the mechanism for the Dieckmann condensation.



R and R' are alkyl groups

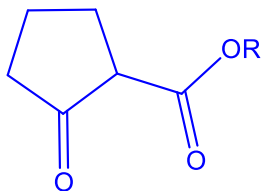
Fig. 3.2

- (i) Suggest the role of $R'O^-$ in the first step of the reaction. [1]
- (ii) Suggest the structure of the organic product formed when the following compound undergoes the Dieckmann condensation. [1]



(i) (Brønsted-Lowry) base / proton acceptor

(ii)



- (c) Under acidic conditions, ketones react with halogens to give substitution at the alpha(α)-carbon. Fig. 3.3 shows the mechanistic steps that occur with propanone as the organic reagent.

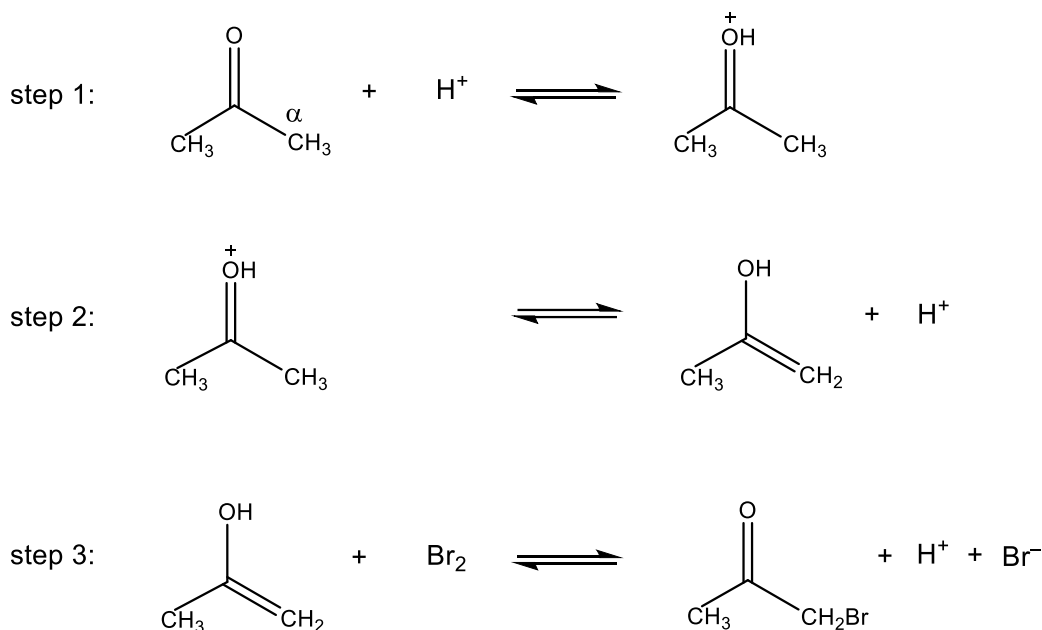


Fig. 3.3

- (i) Write a balanced equation for the overall reaction that occurs when propanone is mixed with acidified Br_2 . [1]
- (ii) Briefly outline how the initial rate of the reaction in (c) could be measured. [2]
- (iii) A student conducted an experiment and determined that the rate equation for the reaction in (c) is $\text{rate} = k[\text{propanone}][\text{H}^+]$, but was unable to conclude whether step 1 or step 2 is the slow step. Suggest why. [1]

(i)

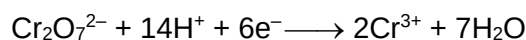
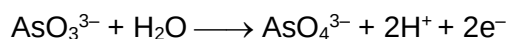
$$\text{CH}_3\text{C}(=\text{O})\text{CH}_3 + \text{Br}_2 \longrightarrow \text{CH}_3\text{C}(=\text{O})\text{CH}_2\text{Br} + \text{H}^+ + \text{Br}^-$$

(ii) Measure the change in colour intensity/concentration of Br_2 /electrical conductivity with time.
Then plot the graph of colour intensity/concentration of Br_2 gas used against time. Determine the gradient at $t = 0$ which is the initial rate.

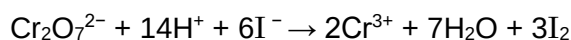
(iii) Regardless of whether step 1 or step 2 is the slow step, the same rate equation is obtained.

- (d) Groundwater contaminated with arsenite, AsO_3^{3-} , presents a grave concern in many developing countries. To protect public health, the World Health Organisation sets a permissible limit for **arsenic**(As) in drinking water; typically at 10 micrograms per dm^3 ($\mu\text{g dm}^{-3}$ or $10^{-6} \text{ g dm}^{-3}$). The level of arsenic toxicity is tested based on the amount of arsenite that is presented in the water source.

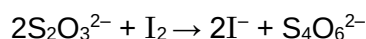
A 100 cm^3 sample of water is drawn from a polluted source. 50.00 cm^3 of $0.00110 \text{ mol dm}^{-3}$ acidified $\text{Cr}_2\text{O}_7^{2-}$ (aq) ions, which is in excess, is added to the solution. The two half-equations for the reaction that occurs are as shown:



An excess of iodide ions is then added to react with the remaining $\text{Cr}_2\text{O}_7^{2-}$, liberating iodine.

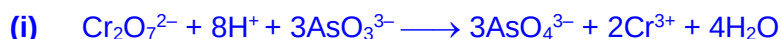


The liberated iodine is then titrated against $0.0150 \text{ mol dm}^{-3}$ $\text{Na}_2\text{S}_2\text{O}_3$.

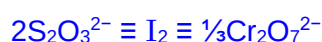


The end-point is reached when 21.20 cm^3 of $\text{Na}_2\text{S}_2\text{O}_3$ has been added.

- Construct the ionic equation for the reaction between arsenite ions, AsO_3^{3-} , and acidified dichromate(VI) ions, $\text{Cr}_2\text{O}_7^{2-}$. [1]
- Determine the amount of arsenite, AsO_3^{3-} , in the 100 cm^3 sample of water. [3]
- Hence calculate how many times the **arsenic** level in the water sample exceeds the permissible limit for drinking water. [2]



(ii) Moles of $\text{S}_2\text{O}_3^{2-} = \frac{21.20}{1000} \times 0.0150 = 3.18 \times 10^{-4} \text{ mol}$



$$\begin{aligned} \text{Moles of } \text{Cr}_2\text{O}_7^{2-} \text{ (unreacted excess)} &= \frac{1}{6} \times (3.18 \times 10^{-4}) \\ &= \underline{5.30 \times 10^{-5} \text{ mol}} \end{aligned}$$

$$\text{Initial moles of } \text{Cr}_2\text{O}_7^{2-} \text{ added} = \frac{50}{1000} \times 0.00110 = 5.50 \times 10^{-5} \text{ mol}$$

$$\begin{aligned} \text{Moles of } \text{Cr}_2\text{O}_7^{2-} \text{ reacted with } \text{AsO}_3^{3-} \\ &= (5.50 \times 10^{-5}) - (5.30 \times 10^{-5}) = \underline{2.00 \times 10^{-6} \text{ mol}} \end{aligned}$$

$$\begin{aligned} \text{Moles of } \text{AsO}_3^{3-} \text{ in } 100 \text{ cm}^3 \text{ sample} &= 3(2.00 \times 10^{-6}) \\ &= \underline{6.00 \times 10^{-6} \text{ mol}} \end{aligned}$$



(iii) Mass of As in 100 cm³ sample = $74.9 \times 6.00 \times 10^{-6}$
= 4.494×10^{-4} g

Mass of As in 1 dm³ = $4.494 \times 10^{-4} \times 10$
= 4.494×10^{-3} g
= 4494×10^{-6} g

It exceeds the permissible limit by = $\frac{4494}{10} = \underline{\underline{449.4 \text{ times}}}$



- (e) Aromatic compound **J** ($C_{10}H_{12}O_3$) gives a positive result with both Tollens' reagent and Fehling's solution. It forms a yellow precipitate when heated with alkaline aqueous iodine and produces white fumes when exposed to $SOCl_2$.

When **J** reacts with concentrated H_2SO_4 under hot conditions, it forms **K** ($C_{10}H_{10}O_2$), which does not exhibit stereoisomerism. Both **J** and **K** decolourise aqueous Br_2 but in different mole ratios, as shown in Table 3.1.

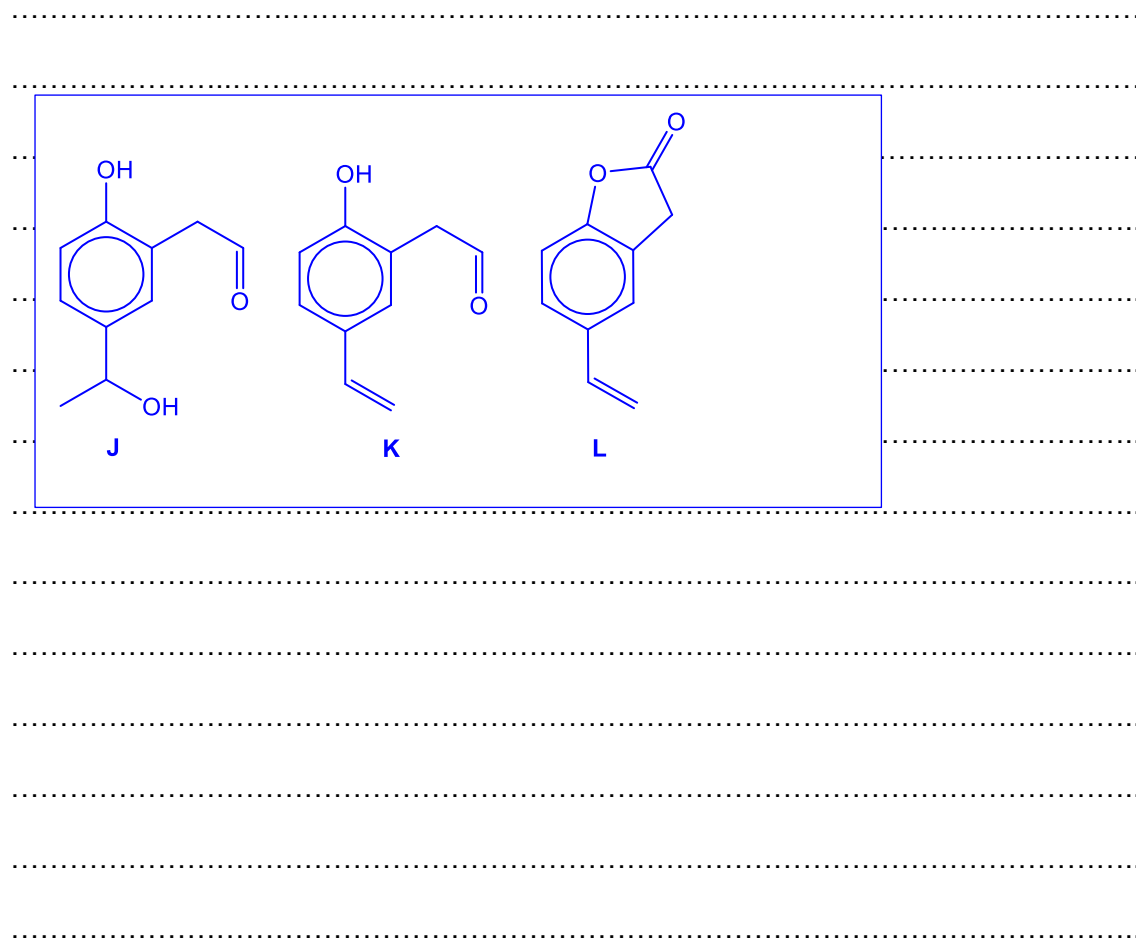
Table 3.1

compound	moles of $Br_2(aq)$
J	1
K	2

K, when heated with acidified potassium dichromate and subsequently treated with PCl_5 , forms **L** ($C_{10}H_8O_2$), which emits a sweet smell. Both **J** and **K** form a violet complex with neutral $FeCl_3$ but **L** does not.

Give the structure of compounds **J**, **K** and **L**.

[3]



[Total: 23]

Section B

Answer **one** question from this section.

- 4 Aromatic compounds serve as important building blocks in organic synthesis. They are widely used as starting materials to produce pharmaceuticals, dyes, fragrances, and many other chemicals.

- (a) Both methylbenzene and phenol form nitro compounds with nitric acid under different conditions as shown in Fig. 4.1.

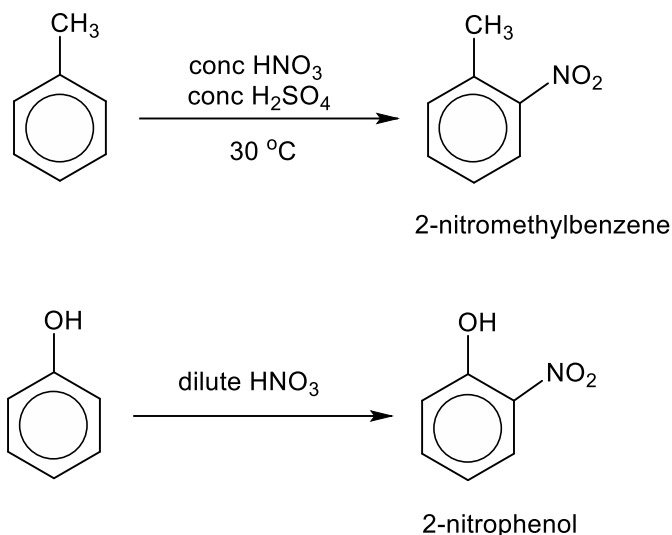


Fig. 4.1

- (i) Explain why different conditions are needed for the formation of 2-nitrophenol and 2-nitromethylbenzene as shown in Fig. 4.1. [2]
- (ii) Hence suggest the reagents and conditions required for a similar reaction of benzoic acid to form the corresponding nitro compound. [1]

- (i)
- The lone pair of electrons on O atom in phenol delocalised into the benzene ring.
 - increasing the electron density and
 - hence increasing reactivity towards electrophile.
 - Hence, the $-\text{OH}$ group in phenol is more activating than $-\text{CH}_3$ in methylbenzene.
 - Since phenol is more susceptible to electrophilic substitution, conc. H_2SO_4 is not required to generate the electrophile.
- (ii) conc HNO_3 , conc H_2SO_4 , heat (or $55\sim 100^\circ\text{C}$)

- (b) Phenol is usually used for the synthesis of *paracetamol* which is a medication for treatment of pain and fever. *Paracetamol* can be prepared from phenol via a three-step synthesis as shown in Fig. 4.2.

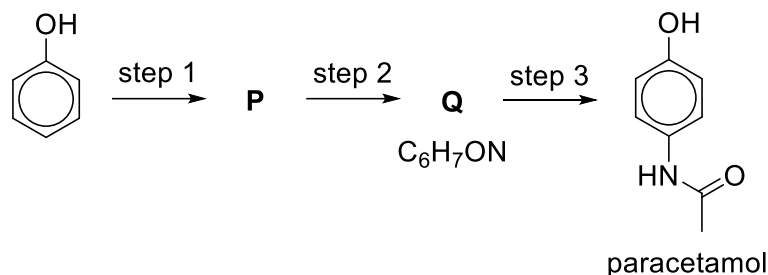


Fig. 4.2

- (i) Suggest the structures of compounds **P** and **Q**, and the reagents and conditions for steps 1 and 2. [3]
- (ii) A student suggested that ethanoyl chloride, CH_3COCl , could be used as a reagent for step 3. However, this reagent may form another compound **R** instead of paracetamol.

Suggest the structure of compound **R**. [1]

Aspirin can also be used for treatment of pain and fever. The industrial production of aspirin involves the conversion of phenol to salicylic acid by the Kolbe-Schmitt reaction. In the reaction, phenol is strongly heated with CO_2 and NaOH as shown in Fig. 4.3.

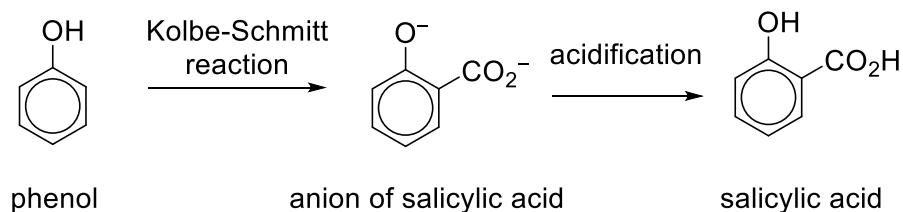
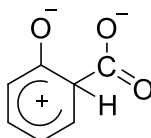


Fig. 4.3

One suggested mechanism for the Kolbe-Schmitt reaction is similar to an electrophilic substitution reaction. The three steps of the mechanism are described below.

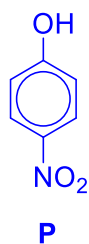
- Step 1: Phenol undergoes acid-base reaction with hydroxide ion to form a phenoxide ion.
- Step 2: CO_2 acts as an electrophile to react with phenoxide ion to form the following intermediate.



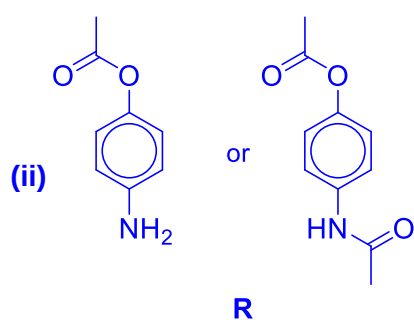
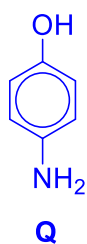
Step 3: The hydroxide ion reacts with the intermediate to form the anion of salicylic acid.

- (iii) Suggest the mechanism for the reaction described above, showing all charges, dipoles and using curly arrows to show the movement of electron pairs in each step. [3]

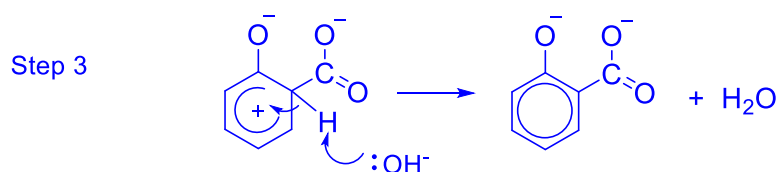
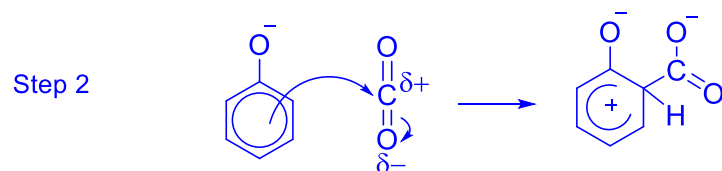
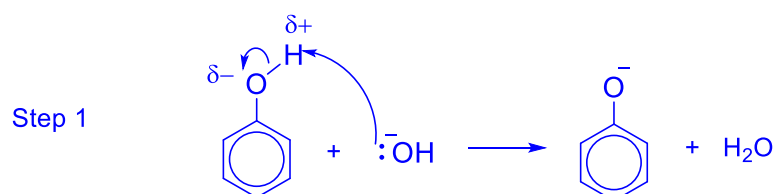
(i) step 1: dilute nitric acid



step 2: Sn and concentrated HCl, heat followed by NaOH(aq)



(iii)



(c) The reactions of the Period 3 chlorides with water vary across the Period.

- (i) Describe the reactions of MgCl_2 and PCl_5 with water, suggesting the pH of the resulting solutions. Explain the reaction of these chlorides in water based on their structure and bonding. Write equations for all reactions that occur. [3]
- (ii) Both CCl_4 and SiCl_4 are Group 14 chlorides. SiCl_4 fumes in moist air due to a vigorous reaction with water. However, CCl_4 does not react with water. Suggest a reason for the unreactivity of CCl_4 towards water. [1]

(i) MgCl_2

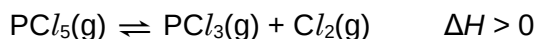
- MgCl_2 has a giant ionic lattice structure and strong electrostatic attraction/ionic bonds between Mg^{2+} and Cl^- ions.
- MgCl_2 dissolves in water readily
- to form an acidic solution of pH 6.5.
- MgCl_2 undergoes hydration and partial hydrolysis of $\text{Mg}^{2+}(\text{aq})$ due to the polarisation of water molecules by the Mg^{2+} ion.
- $\text{MgCl}_2(\text{s}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow [\text{Mg}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq})$
- $[\text{Mg}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons [\text{Mg}(\text{H}_2\text{O})_5\text{OH}]^+(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$

PCl_5

- PCl_5 has a simple molecular structure and
- undergoes complete hydrolysis in water.
- It forms white fumes of HCl and
- an acidic solution of pH 2.
- $\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl}(\text{aq})$

- (ii) CCl_4 is unreactive towards water as carbon does not have energetically accessible empty 3d orbitals to accept the lone pair of electrons/expand its octet structure from water molecules.

- (d) Upon heating, PCl_5 undergoes decomposition to form PCl_3 and Cl_2 as shown.



A mixture of 0.0625 mol of PCl_5 and 0.0705 mol of Cl_2 were placed into a 2.0 dm^3 sealed vessel at 250°C . After dynamic equilibrium was established, the total pressure in the vessel was 3.25 atm. You may assume that the gases behave ideally under the stated conditions.

- (i) By calculating the total amount of gases in the vessel at equilibrium or otherwise, calculate the amount of PCl_3 formed at equilibrium at 250°C . [2]
- (ii) Calculate the equilibrium constant, K_p at 250°C , stating the units of K_p clearly. [2]
- (iii) State and explain how the K_p value would change when the temperature of the vessel is increased. [2]

(i)

$$pV = nRT$$

$$(3.25 \times 101325) \times (2 \times 10^{-3}) = (n) \times 8.31 \times (273 + 250)$$

$$\underline{n = 0.1515 \text{ mol}} \text{ (total amt of gases)}$$

Let the amount of PCl_3 formed at equilibrium be y .

	PCl_5	$\rightleftharpoons$	PCl_3	+	Cl_2
Initial /mol	0.0625		0		0.0705
Change /mol	$-y$		$+y$		$+y$
Eqm /mol	$0.0625 - y$		y		$0.0705 + y$

$$\text{Total amount of gas} = 0.0625 - y + y + 0.0705 + y$$

$$0.1515 = 0.133 + y$$

$$y = 0.0185$$

$$\text{Amount of } \text{PCl}_3 \text{ formed at equilibrium} = \underline{0.0185 \text{ mol}}$$

(ii) Eqm partial pressure of $\text{PCl}_5 = \frac{0.0625 - 0.0185}{0.1515} \times 3.25 = \underline{0.944 \text{ atm}}$

$$\text{Eqm partial pressure of } \text{Cl}_2 = \frac{0.0705 + 0.0185}{0.1515} \times 3.25 = \underline{1.91 \text{ atm}}$$

$$\text{Eqm partial pressure of } \text{PCl}_3 = \frac{0.0185}{0.1515} \times 3.25 = \underline{0.397 \text{ atm}}$$

$$K_p = \frac{P_{\text{PCl}_3} P_{\text{Cl}_2}}{P_{\text{PCl}_5}} = \frac{0.397 \times 1.91}{0.944} = \underline{0.803 \text{ atm}}$$

- (iii) By LCP, when temperature is increased, position of equilibrium will shift right towards the endothermic reaction in order to absorb some heat. Hence the value of K_p increases.

[2]



- 5 (a) The Group 2 carbonates are widely used in the construction and agriculture industry.

On heating, the Group 2 carbonates decompose as shown:



- (i) Describe and explain the trend in the thermal stabilities of the Group 2 carbonates. [2]
- (ii) The graph in Fig. 5.1 represents the change in mass that occurs when 1.0 g of powdered calcium carbonate, CaCO_3 , is heated at a certain temperature.

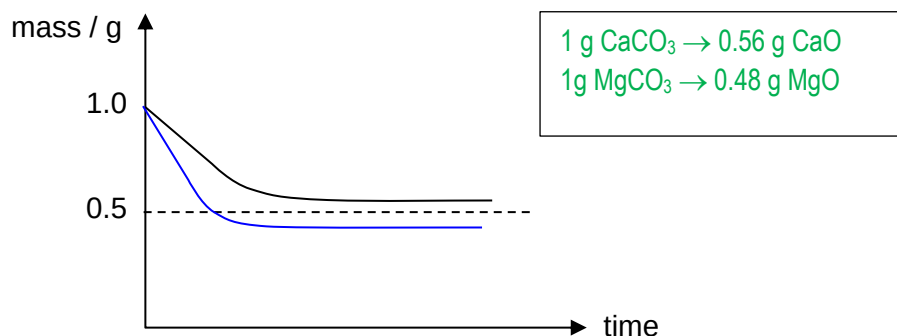


Fig. 5.1

Sketch, on the same axis, a graph that would be obtained by heating 1.00 g of magnesium carbonate, MgCO_3 , at the same temperature.

Your answer should illustrate clearly the relative ease of thermal decomposition of the two carbonates and the relative mass of residue formed.

Include any relevant calculations.

[M_r : MgCO_3 , 84.3 ; CaCO_3 , 100.1]

[2]

- (i)
- Down Group 2, charge density of cations decreases.
 - Polarising power of the cation decreases / ability of cations to distort the anion decreases.
 - C–O bond in the anion is weakened to a lesser extent / less energy required to break C–O bond in anion.
 - Thermal stability of Group 2 carbonates increases down the Group.
- (ii)
- $\text{MCO}_3 \rightarrow \text{MO} + \text{CO}_2$
- 1 g CaCO_3 gives $\frac{1}{100.1} \times 56.1 = 0.560$ g of CaO residue.
- 1 g MgCO_3 gives $\frac{1}{84.3} \times 40.3 = 0.478$ g of MgO residue

- (b) The biological process of forming proteins from amino acids is enzyme catalysed.

Graph 1 in Fig. 5.2 shows how the rate of an enzyme catalysed reaction varies with substrate(reactant) concentration for a given enzyme concentration.

Graph 2 in Fig. 5.2 shows how the rate of an enzyme catalysed reaction varies with enzyme concentration when the enzyme concentration is significantly lower than the substrate concentration.

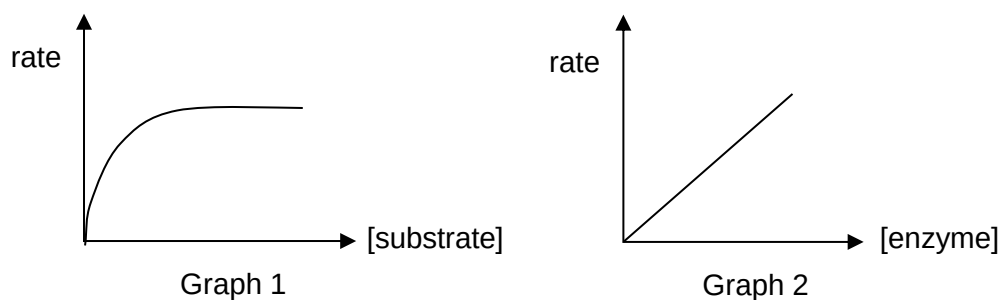


Fig. 5.2

- (i) Explain why Graph 1 in Fig. 5.2 levels off at high [substrate]. [1]
- (ii) State the relationship between reaction rate and enzyme concentration as shown in Graph 2. Suggest a reason for this. [1]

- (i) Graph 1 levels off at high [substrate] as the active sites on the enzyme are saturated/completely occupied.
- (ii) Graph 2 shows that reaction rate is directly proportional to [enzyme] as increasing [enzyme] provides more active sites for more substrate to be reacted.

(c) The Strecker Synthesis is a laboratory method of forming amino acids from aldehydes.

Fig. 5.3 shows an overview of this multi-step synthesis that involves forming an α -aminonitrile, which is then subsequently carefully hydrolysed to form the amino acid.

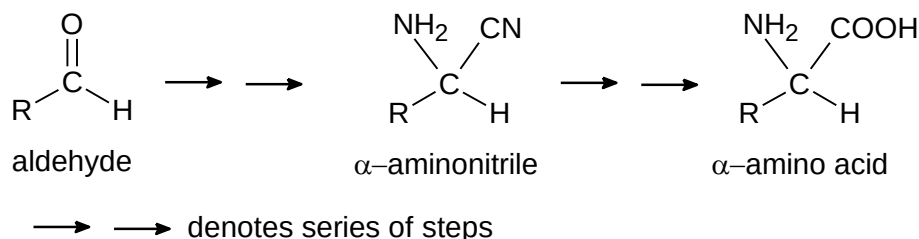
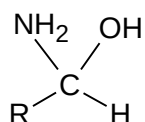


Fig. 5.3

(i) In the conversion of the aldehyde to an α -aminonitrile shown in Fig. 5.3, the first 2 steps involve a nucleophilic addition which occurs as described:

- nucleophilic attack of ammonia on the carbonyl carbon to yield a dipolar ion consisting of both a positive and a negative charge;
- internal proton transfer to form a neutral molecule with the structure below.

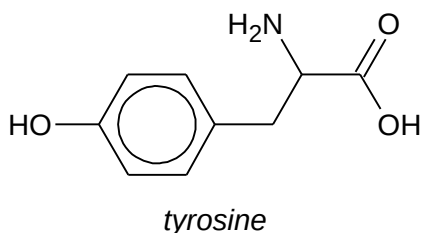


Draw the mechanism described in the 2 steps above, showing clearly the structure of the dipolar ion. Include curly arrows and relevant dipoles in your answer. [3]

(ii) By considering the information given in (c)(i), explain why the product formed is likely a racemic mixture. [1]

(iii) Suggest why the α -aminonitrile had to be “carefully hydrolysed” to form the amino acid. [1]

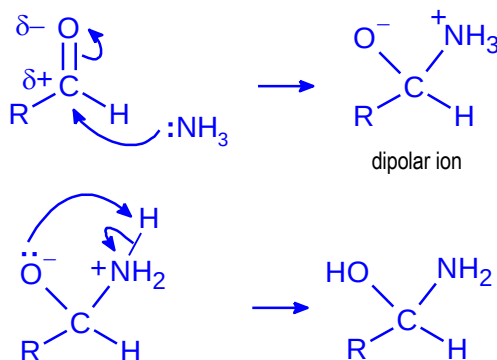
(iv) Draw the **skeletal** formula of the aldehyde that can be used to form the amino acid *tyrosine* via the Strecker Synthesis. [1]



(v) Tyrosine has three pK_a values associated with it: 2.2, 9.1 and 10.0.

Given that $pK_a = 10.0$ is for the phenolic R group, draw the structure of the predominant species of tyrosine at pH 7. [1]

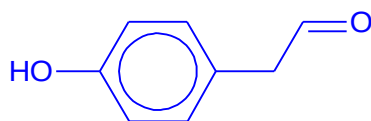
(i)



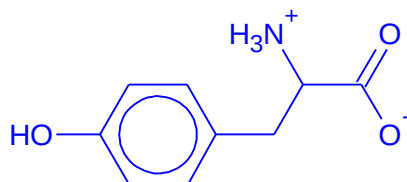
(ii) Ammonia/nucleophile attacks the trigonal planar sp^2 hybridised electron-deficient(δ^+) carbon from above and below the plane with equal probability, resulting in equal amounts of (+) and (-) enantiomers.

(iii) The hydrolysis had to be controlled so that the $-NH_2$ would not be protonated. (accept any chemically sound answer)

(iv)



(v)



(d) Fig. 5.4 shows a part of the protein structure of human insulin.

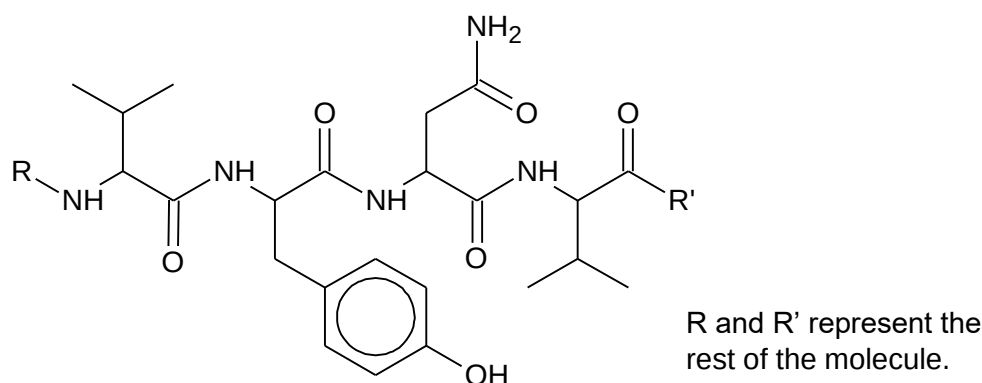


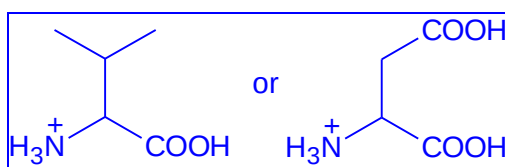
Fig. 5.4

- (i) State the reagents and conditions required to completely hydrolyse a protein molecule into its constituent amino acids. [1]
- (ii) The insulin molecule is hydrolysed using the reagent and conditions given in (i). Draw the structure of a species in the resulting reaction mixture that is formed from one of the amino acids other than tyrosine. [1]

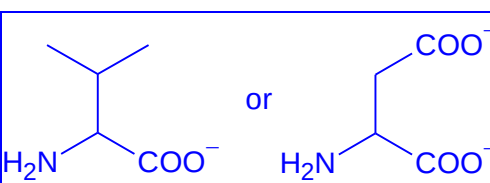
(i) HCl(aq) or H₂SO₄(aq) OR NaOH(aq);
prolonged heating (or heat under reflux)

(ii)

acidic hydrolysis



basic hydrolysis



- (e) Compound **X** has molecular formula $C_7H_{13}NO_2$. On boiling **X** with aqueous sodium hydroxide, ethylamine is formed.

- (i) Ethylamine dissolves readily in dilute acid at room conditions whereas compound **X** does not.

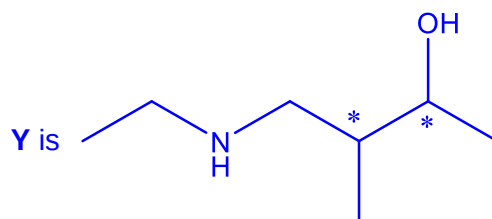
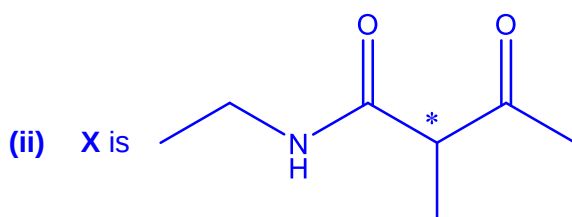
Suggest the nitrogen-containing functional group that could be present in **X**.
Explain the observed difference in acid-base nature of the nitrogen-containing functional groups. [2]

- (ii) Compound **X** is optically active. Compound **Y**, $C_7H_{17}NO$, which is also optically active, is formed when **X** reacts with $LiAlH_4$. On heating with Al_2O_3 , **Y** forms two different compounds which readily decolourise aqueous bromine.

Suggest possible structures for **X** and **Y**, stating the type of reaction that has occurred for each. [3]

- (i) Ethylamine has a lone pair of electrons on N which is available for dative bonding with H^+ . Hence it is basic/undergoes neutralisation with dilute acid.

X contains an amide functional group in which the lone pair on nitrogen is delocalised into the adjacent $C=O$. Hence it is neutral/not basic (or lone pair unavailable to accept H^+).



X : reduction, **Y** : elimination

[Total: 20]



Additional answer space

If you use the following to complete the answer to any question, the question number must be clearly shown.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.